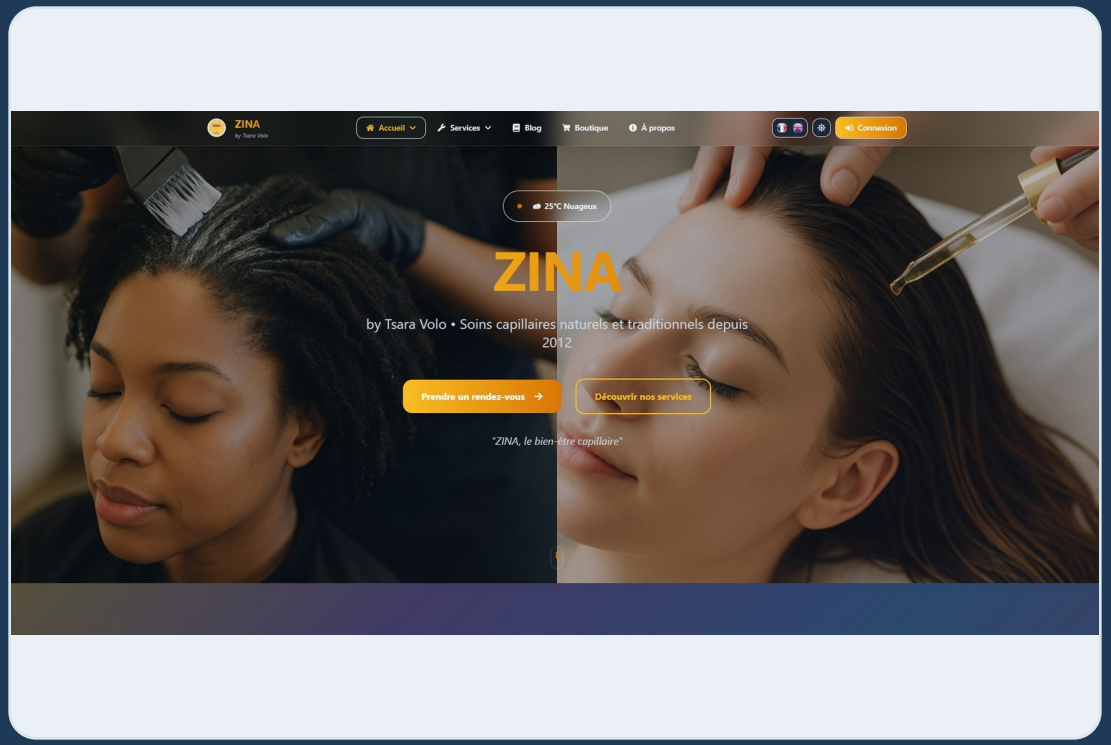
TECHNICAL CASE STUDY · ENGLISH EDITION

ZINA by Tsara Volo

Customer relationship-focused web platform.



LICENCE FINAL-YEAR PROJECT · 2025

A public website, customer spaces and internal customer relationship management.

TECH STACK

React / Vite · Spring Boot · PostgreSQL · Internal CRM

Presented by: RANDRIAMANANTENA Fitahiana Lafatra

Project overview

Digitizing a store, services and customer relations.

SETTING · ACADEMIC INTERNSHIP

DA2I Licence final-year project, based on the needs of a hair-care business.

CHALLENGE · FRAGMENTED JOURNEYS

The store, appointments and customer relationship history needed to be better connected.

SOLUTION · INTEGRATED PLATFORM

Public, customer and administrator areas, with an internal CRM and activity monitoring.

TECHNOLOGIES · REACT + SPRING BOOT

Modern frontend, Java backend and PostgreSQL relational persistence.

Bringing together four business areas

Product logic organized around the customer experience.

01 · COMMERCE

Catalog, shopping cart, orders and tracking.

02 · SERVICES

Appointment bookings, treatments and hair diagnostics.

03 · CUSTOMER RELATIONS

History, communication and loyalty.

04 · MANAGEMENT

Internal CRM, content and management views.

Three areas for one customer journey

User profiles shape navigation and access levels.

PUBLIC · EXPLORE

Browse the website, services, products and articles.

CUSTOMER · INTERACT

Prepare purchases, consult requests and manage the personal profile.

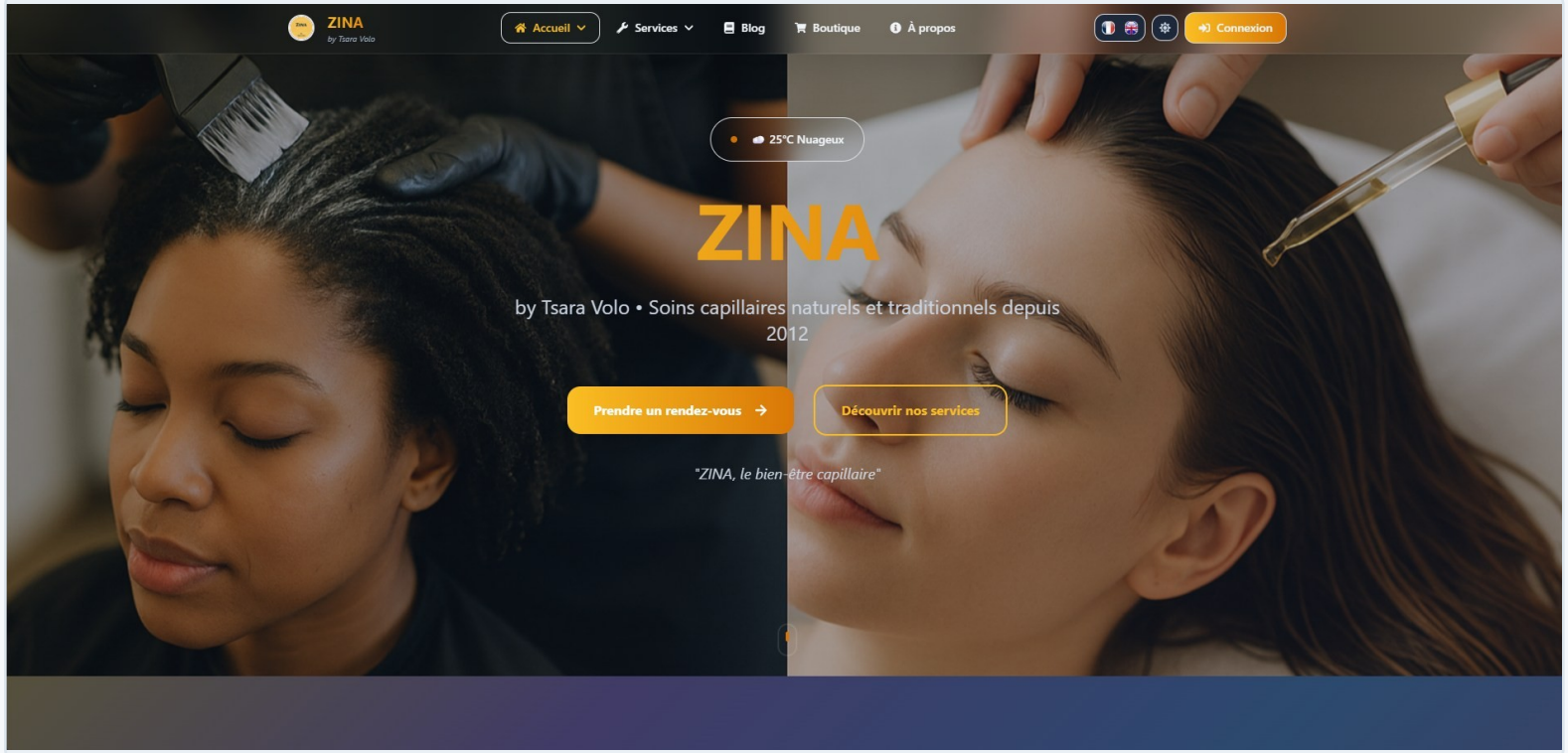
ADMIN · ORGANIZE

Manage content, appointments and customer relationships.

RULE · Permissions must be checked by the API, not only in the frontend.

ZINA home page and visual identity

Public interface · original project screenshot.



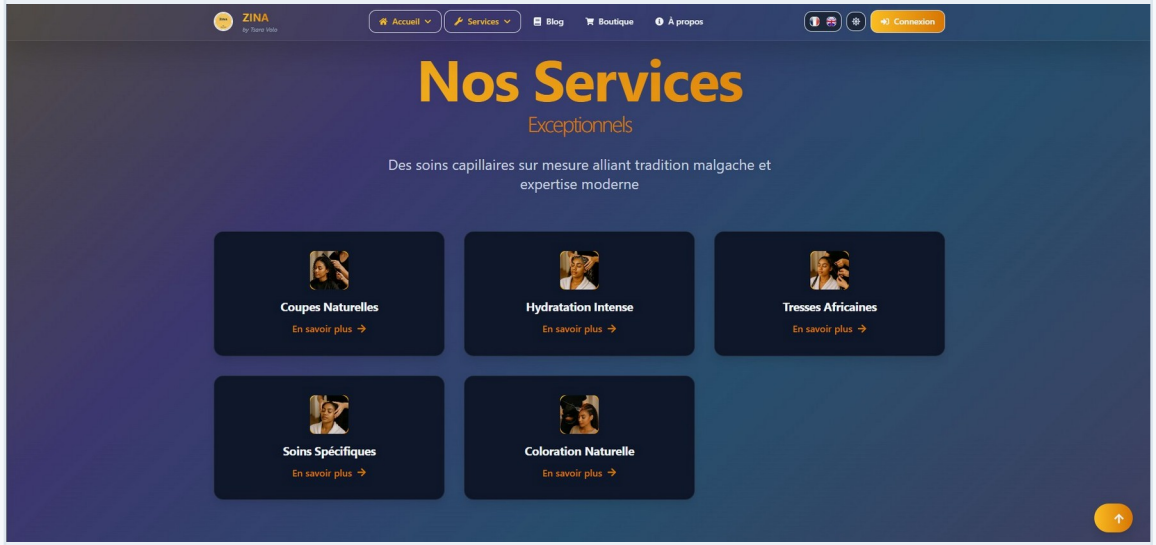
Original screenshot from the supplied presentation.

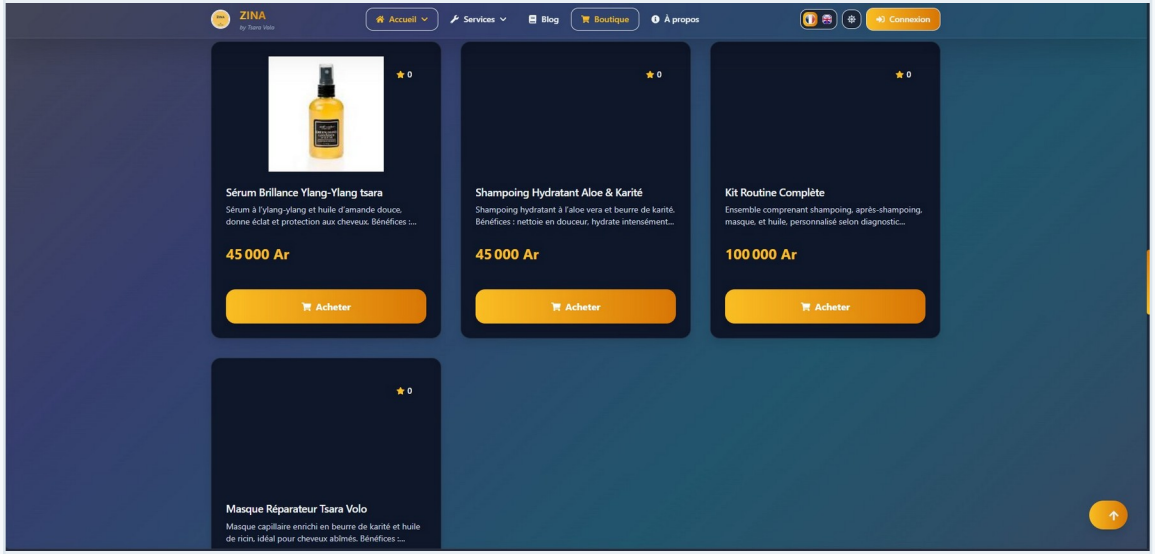
WHAT THE SCREENSHOT SHOWS

Hair-care-themed homepage; navigation to services, shop and blog; French/English content and visual personalization.

Services and online shop

Service and product browsing interfaces.





SERVICE CATALOG

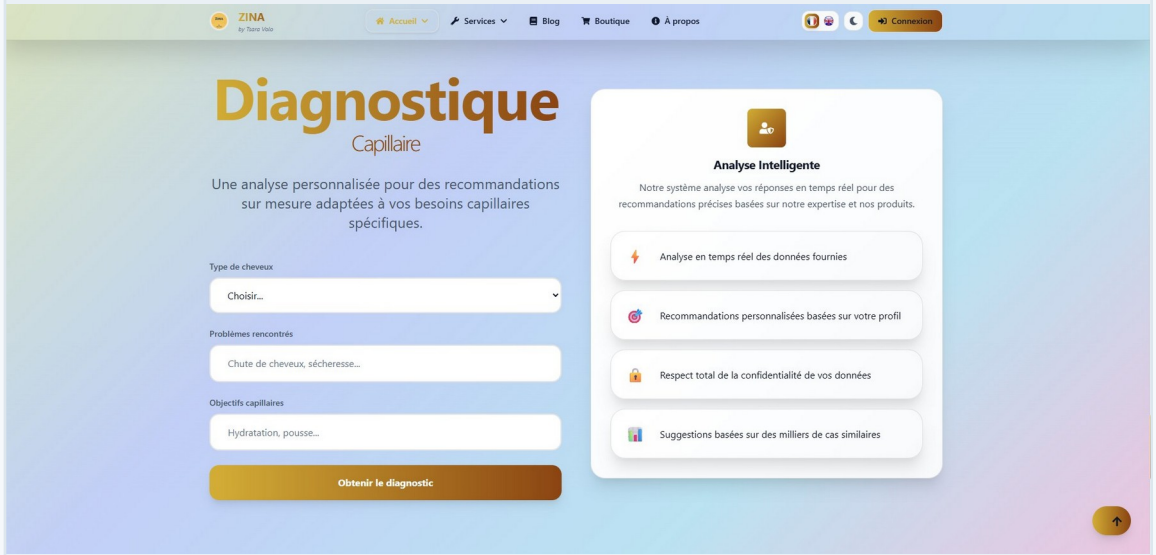
Presentation of treatments, services and their related user journeys.

PRODUCT CATALOG

Product pages, prices in Malagasy ariary and purchase actions.

Hair diagnostics and editorial content

Public information and personalized service discovery.





HAIR DIAGNOSTICS

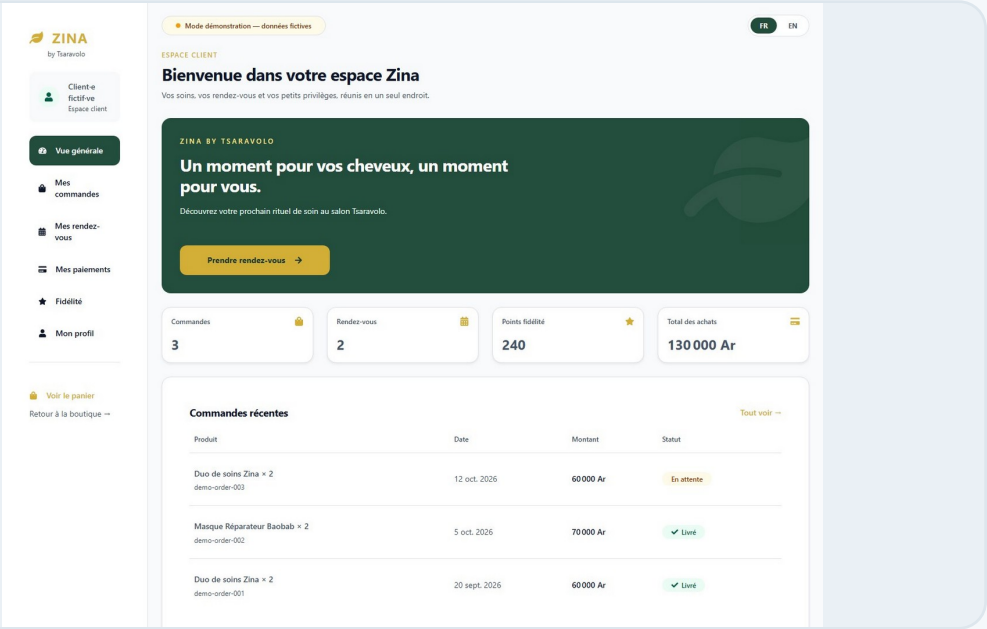
Hair-profile form and guidance towards recommendations.

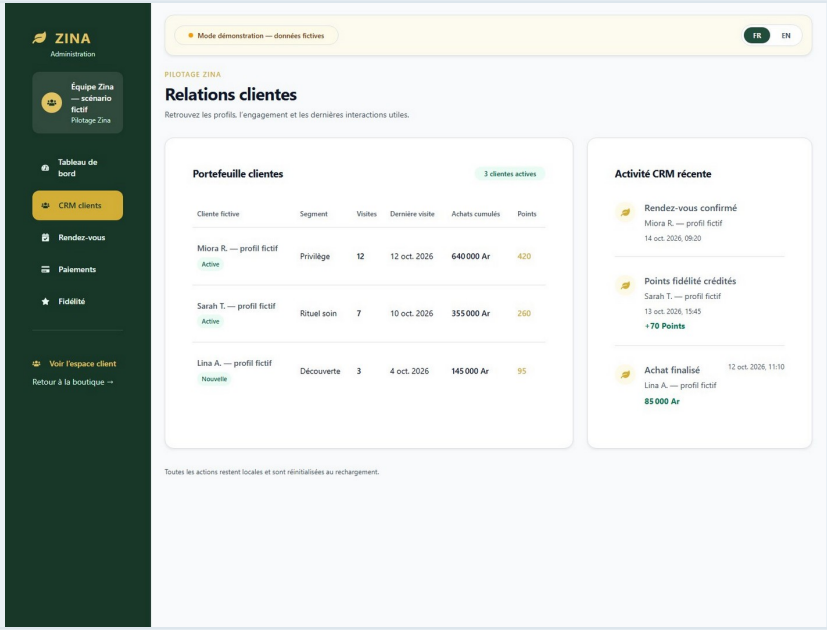
PUBLIC BLOG

Articles intended to inform users and promote care services.

Customer and administration spaces

The screenshots use fictional data - these are local demonstrations.





CUSTOMER AREA · DEMO

Customer dashboard: orders, appointments and loyalty.

ADMIN CRM · DEMO

Admin CRM: segments, visits, purchases and simulated interactions.

Separate React frontend and Java backend

Layered architecture described in the project dissertation.

01 · CLIENT

React / Vite. Components, forms, navigation and French/English support.

02 · HTTP

REST API. JSON exchanges between the application and the server.

03 · BUSINESS LOGIC

Spring Boot. Business services, secured access and notifications.

04 · SQL

PostgreSQL. Relational model with Spring Data JPA.

CROSS-CUTTING · A documented architecture does not prove that every endpoint is deployed or tested.

Documented data model

Key entities and business relationships in the proposed model.

USERS / ROLES

Accounts and permissions.

PRODUCTS

Products and catalog.

ORDERS / ORDER ITEMS

Orders and product lines.

APPOINTMENTS

Appointments and statuses.

PAYMENTS

References and statuses.

LOYALTIES / HISTORIES

Loyalty and events.

RELATIONSHIPS · users → orders → order_items → products; users → appointments; orders → payments. The dissertation lists 21 target tables, a count not verified in a restored PostgreSQL dump.

From form submission to request tracking

Appointment emails received in certain scenarios appear in the supplied materials.

01 · SELECT

Choose a service and a time slot.

02 · RECORD

Create a request with a status.

03 · PROCESS

Approve, reschedule or cancel, depending on the workflow.

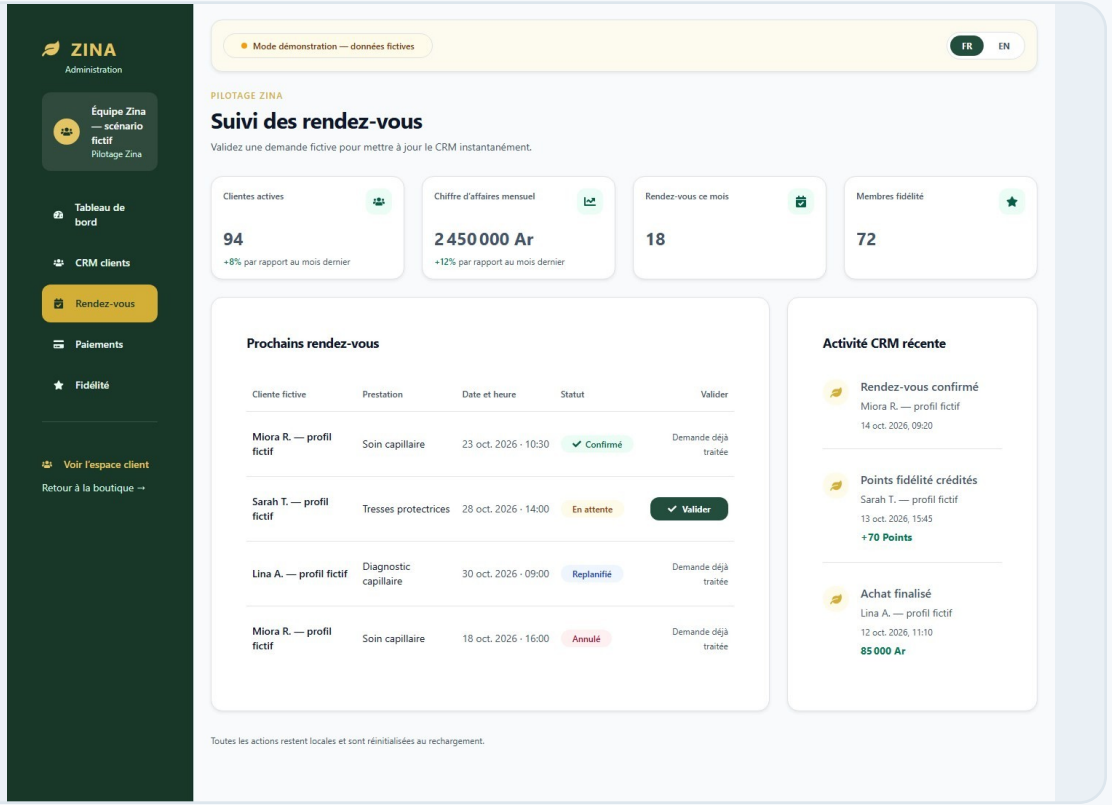
04 · NOTIFY

Inform the customer by email when the scenario provides for it.

NOTE · Screenshots do not certify every email-sending path or production availability.

CRM and appointments: scenario views

The screens illustrate usage logic with no real customer records.



DEMONSTRATION MODE — Fictional records and appointments.

ILLUSTRATED ACTION — An example appointment approval action.

FICTIONAL REPORTS — Illustrative scenario statistics, not real measurements.

LIMITATION — No effective real-world booking is assumed.

Illustrated features and outstanding validation

A presentation faithful to the available evidence.

ILLUSTRATED · INTERFACES & NOTICES

Home page, products, services, diagnostics, blog and sample emails.

SCENARIOS · SIMULATED WORKFLOWS

CRM, loyalty, shopping cart and appointments with clearly marked fictional data.

EVOLUTION · REAL INTEGRATIONS

End-to-end tests, API security and automated payment have not been certified.

TRANSPARENCY · No trained AI model or automated Mobile Money settlement is claimed without proof.

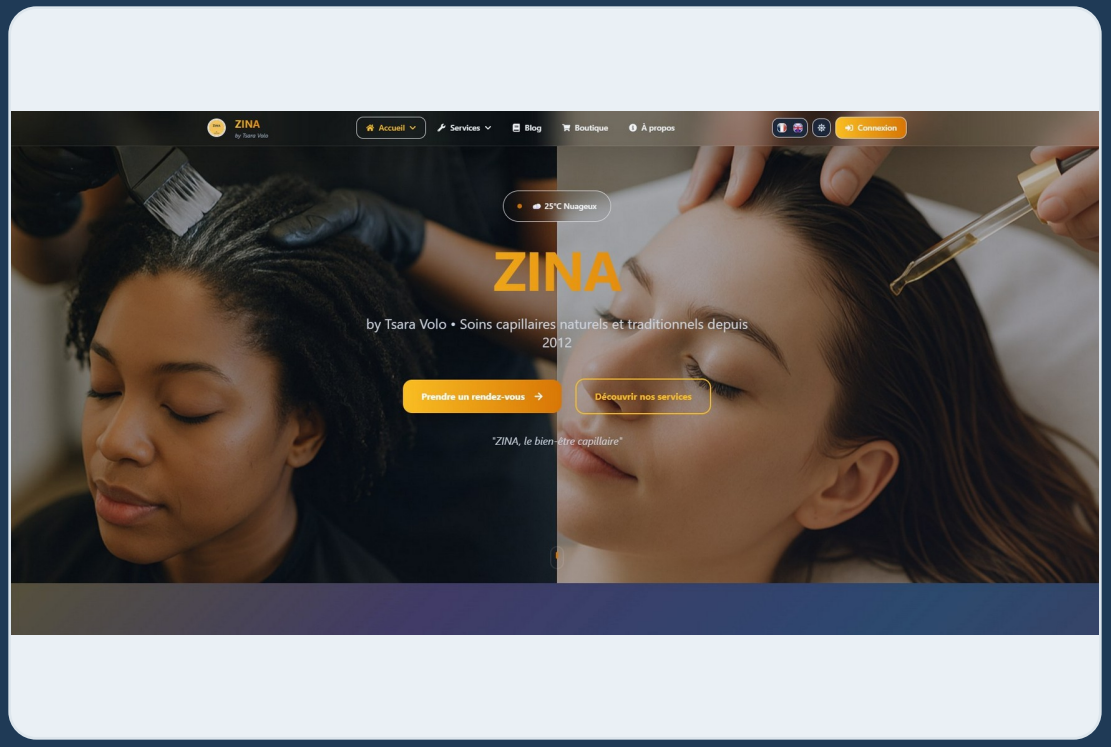
TECHNICAL CASE STUDY · ENGLISH EDITION

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A full-stack project linking commerce, services and customer relationships through React and Java.

TECHNOLOGIES

React / Vite · Spring Boot · PostgreSQL · Internal CRM



Screenshots and architecture based on project materials. No production deployment is claimed.

Presented by: RANDRIAMANANTENA Fitahiana Lafatra